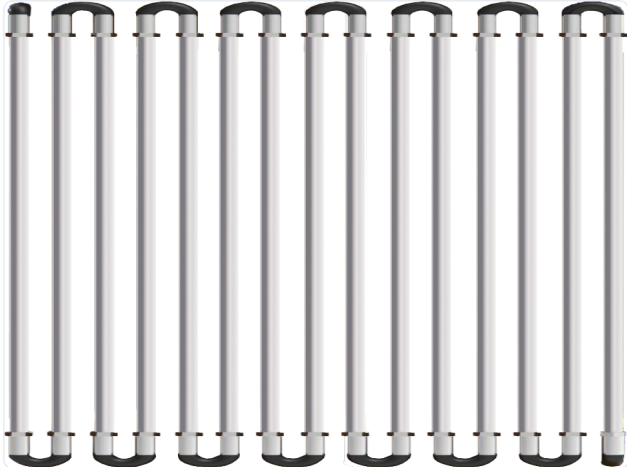


INTRODUCTION

The Flexible Inclinator, also known as an Array Displacement Meter, is a multi-section, flexible three-dimensional measurement system for geotechnical deformation monitoring. Its modular sensing units can be connected in series and assembled on site according to borehole depth, making the system reusable, easy to transport, and convenient to maintain.

The system is designed for static deformation monitoring in rock and soil engineering, including deep horizontal displacement, settlement monitoring, tunnel convergence, foundation pit monitoring, embankment settlement, and underground structure convergence.



KEY FEATURES

MODULAR FLEXIBLE DESIGN

Multi-section sensing units can be freely assembled to match different borehole depths and reused across projects.

TEMPERATURE COMPENSATION

The MEMS system uses temperature-zone compensation to reduce temperature drift and improve data stability.

OFFSET AND TORSION CORRECTION

Installation direction offset and inclinometer casing torsion can be corrected to improve monitoring accuracy.

EASY DEPLOYMENT AND MAINTENANCE

Standardized sections support batch stocking, quick replacement, convenient transport, and fast field maintenance.

THREE-DIMENSIONAL MEASUREMENT

Measures X, Y, and Z axis deformation to support deep displacement, settlement, and convergence monitoring.

FACTORY CALIBRATED NODES

Each node is independently calibrated with a high-precision automatic calibration system, followed by overall calibration after assembly.

ONLINE TRANSMISSION

Supports 4G full-network communication and serial communication for real-time data acquisition, transmission, analysis, and download.

PRINCIPLES, INSTALLATION, DATA & CASES

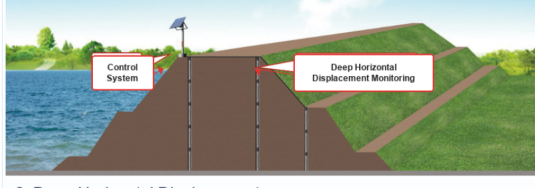
The system uses MEMS accelerometer sensing units to measure acceleration changes along different axes. These changes are converted into angle variations relative to the direction of gravity, and the displacement of each node is calculated from the angle change. Multiple sensing sections are connected in series to form a flexible measuring chain. The total length can be assembled on site based on project depth requirements. After installation, data can be collected continuously and transmitted online through 4G full-network communication or serial communication for real-time cloud analysis.

INSTALLATION

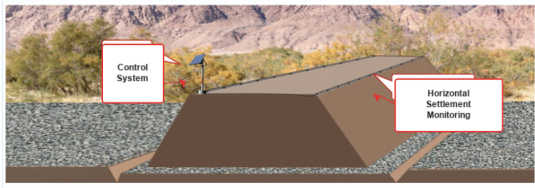
Typical Monitoring Layouts



1. Tunnel Convergence



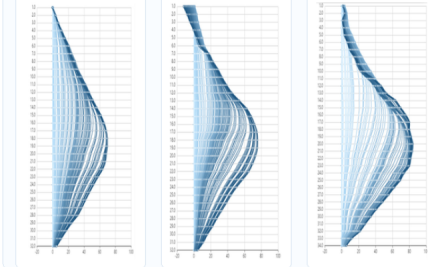
2. Deep Horizontal Displacement



3. Horizontal Settlement

CASE 1

Rail Transit Foundation Pit: Multi-Point Data



Monitoring period
Apr. 2024 to present

Depths
32 m, 34 m, 32 m

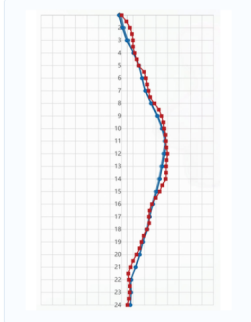
Sampling interval
1 hour

Power
220 V AC to 12 V

Result
Daily displacement changes align with field conditions; manual check data are broadly consistent.

CASE 2

Adjacent Borehole Comparison



Test Site
A metro foundation pit during rail transit construction. Two adjacent inclinometer boreholes were located 20 cm apart.

Comparison Method
A manual sliding inclinometer measured deep horizontal displacement in Borehole 1, while the flexible inclinometer measured Borehole 2.

Conclusion
Over a six-month displacement time-series comparison, both methods showed highly consistent monitoring results. The A-direction deformation trends were fully aligned.



Slope



Railway track



Bridge



Subway

TECHNICAL SPECIFICATIONS

Parameter	Specification
Function type	Array displacement meter / Flexible inclinometer
Working method	Self-developed MEMS
Measurement direction	3D measurement (X, Y, Z axes)
Angle range	0–360°
Repeatability	±0.002°
Angle resolution	1′
Displacement resolution	0.01 mm/m at 1m gauge length
System stability	Better than 0.3 mm / 30 m
Measurement accuracy	0.001%FS; 0.01 mm / 500 mm
Radial compression resistance	Greater than 10 ⁴ N
Diameter specification	18 mm body, 25 mm maximum; segmented joint diameter 29 mm when spliced
Connecting rod coaxiality	0.5 mm
Weight	≤0.6 kg/m
Material	Ultra-high-strength metal material
Temperature measurement accuracy	Better than 0.1°
Torsion correction accuracy	Less than 0.5°
Temperature recording	Real-time temperature recording for each section
Acquisition frequency	10 kHz (10 times per second)
Electrical power consumption	DC 12 V, 3.2 mA/node
Tensile guarantee	Greater than 10 ⁴ N
Waterproof guarantee	300 m, IP68 / 5 MPa; 8 MPa available by project configuration
Working environment	Passed HAST test
Magnetic field interference	Passed EMC test
Electric field interference	Passed EMC test
Deformation trajectory	Real-time playback of deformation trajectory
Length options	Single section 0.3 m, 0.5 m, or 1 m optional
Maximum bending angle at section joint	180°
Assembly method	Free assembly to any length
Segment length	Customizable/0.3 m / 0.5 m / 1 m options